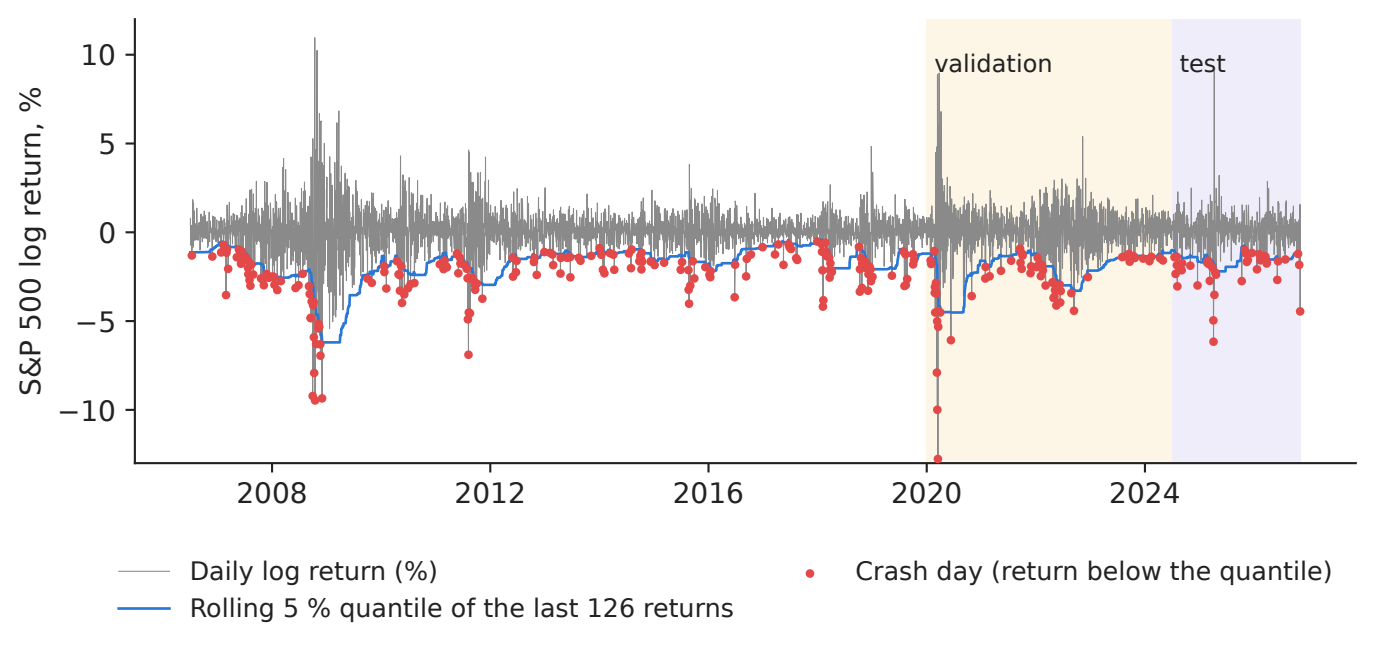

S&P 500 log return, %
10
5
0
−5
−10
validation
test
2008
2012
2016
2020
2024
Daily log return (%)
Rolling 5 % quantile of the last 126 returns
Crash day (return below the quantile)